

● RAG Learning

1.2 Algebra

Mathematics B (MEI) — OCR B — A-Level (H640) — Pure Mathematics — spec refs a1–a15

What this topic covers. The algebraic toolkit the rest of the A-Level runs on: surds and indices, quadratics and the discriminant, simultaneous equations, inequalities and set notation, proportion, and partial fractions.

1. Algebraic language

MEI expects the vocabulary to be used precisely (spec ref a1).

Term	Meaning
Expression	A collection of terms, with no equals sign: $3x^2 - 5x$.
Equation	Two expressions set equal, true only for particular values: $x^2 = 9$.
Identity	True for every value; written with $\equiv$: $(x + 1)^2 \equiv x^2 + 2x + 1$.
Function	A rule assigning exactly one output to each input.
Coefficient	The numerical multiplier of a term: in $4x^3$, it is 4.
Term	A single part of an expression, separated by $+$ or $-$.
Variable / unknown	A quantity that varies / a fixed value to be found.

2. Surds

A **surd** is an irrational root left in exact form. Working in surds keeps answers exact; decimals introduce rounding error.

The rules

$$\sqrt{ab} = \sqrt{a} \sqrt{b}, \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}, \quad (\sqrt{a})^2 = a$$

Simplify by extracting the largest square factor:

$$\sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}, \quad \sqrt{8} + \sqrt{2} = 2\sqrt{2} + \sqrt{2} = 3\sqrt{2}$$

$\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$. Test it: $\sqrt{9+16} = 5$, but $\sqrt{9} + \sqrt{16} = 7$. The product rule holds; the sum rule does not.

Rationalising the denominator

For a single surd, multiply top and bottom by that surd:

$$\frac{1}{\sqrt{5}} = \frac{1}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{5}}{5}$$

For a denominator $a + \sqrt{b}$, multiply by its **conjugate** $a - \sqrt{b}$, using the difference of two squares to clear the surd:

Worked example. Rationalise $\frac{1}{3 + \sqrt{5}}$.

$$\frac{1}{3 + \sqrt{5}} \times \frac{3 - \sqrt{5}}{3 - \sqrt{5}} = \frac{3 - \sqrt{5}}{9 - 5} = \frac{3 - \sqrt{5}}{4}$$

Worked example. Rationalise $\frac{\sqrt{7} - \sqrt{3}}{\sqrt{7} + \sqrt{3}}$.

$$\frac{(\sqrt{7} - \sqrt{3})^2}{(\sqrt{7} + \sqrt{3})(\sqrt{7} - \sqrt{3})} = \frac{7 - 2\sqrt{21} + 3}{7 - 3} = \frac{10 - 2\sqrt{21}}{4} = \frac{5 - \sqrt{21}}{2}$$

3. Indices

Law	Example
$x^a \times x^b = x^{a+b}$	$x^3 x^4 = x^7$
$x^a \div x^b = x^{a-b}$	$x^5 \div x^2 = x^3$
$(x^a)^b = x^{ab}$	$(x^2)^3 = x^6$
$x^0 = 1$ (for $x \neq 0$)	$7^0 = 1$
$x^{-a} = \frac{1}{x^a}$	$2^{-3} = \frac{1}{8}$
$x^{1/n} = \sqrt[n]{x}$	$8^{1/3} = 2$
$x^{m/n} = (\sqrt[n]{x})^m$	$8^{2/3} = 2^2 = 4$

For $x^{m/n}$, take the **root first**, then the power — the numbers stay small. $8^{2/3}$ as $(\sqrt[3]{8})^2 = 4$ is far easier than $\sqrt[3]{64}$.

4. Quadratic equations

Three routes, plus the graphical picture (spec refs a2, a6).

Factorising

$$x^2 - 5x + 6 = 0 \implies (x - 2)(x - 3) = 0 \implies x = 2 \text{ or } x = 3$$

Completing the square

$$x^2 + 6x + 2 = (x + 3)^2 - 9 + 2 = (x + 3)^2 - 7$$

This form reads off the vertex directly: the minimum of $y = (x + 3)^2 - 7$ is at $(-3, -7)$.

The formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Quadratics in disguise

Worked example. Solve $2^{2x} - 5 \cdot 2^x + 4 = 0$.

Let $u = 2^x$. Since $2^{2x} = (2^x)^2 = u^2$:

$$u^2 - 5u + 4 = 0 \implies (u - 1)(u - 4) = 0 \implies u = 1 \text{ or } u = 4$$

So $2^x = 1$ gives $x = 0$, and $2^x = 4$ gives $x = 2$.

Worked example. Solve $x^4 - 5x^2 + 4 = 0$.

Let $u = x^2$: $u^2 - 5u + 4 = 0$, so $u = 1$ or $u = 4$, giving $x = \pm 1$ or $x = \pm 2$ — four roots.

5. The discriminant

For $ax^2 + bx + c = 0$, the **discriminant** is $\Delta = b^2 - 4ac$ (spec ref a3).

Discriminant	Roots	Graph and the x -axis
$b^2 - 4ac > 0$	Two distinct real roots	Crosses twice
$b^2 - 4ac = 0$	One repeated root	Touches (tangent)
$b^2 - 4ac < 0$	No real roots	Never meets it

Worked example. Find the values of k for which $kx^2 + 4x + k = 0$ has a repeated root.

$$b^2 - 4ac = 16 - 4k^2 = 0 \implies k^2 = 4 \implies k = \pm 2$$

Worked example. Find the values of k for which $kx^2 + (k + 3)x + 4 = 0$ has no real roots.

$$(k + 3)^2 - 16k < 0$$

$$k^2 + 6k + 9 - 16k < 0 \implies k^2 - 10k + 9 < 0 \implies (k - 1)(k - 9) < 0$$

A negative product means k lies between the roots: $1 < k < 9$.

"Two distinct roots" means > 0 ; "two real roots" or "real roots" usually allows the repeated case, so ≥ 0 . Read the wording — this distinction decides whether your final answer uses $<$ or $\leq$.

Tangency

A line meets a curve where their equations are equal. Substituting and demanding $\Delta = 0$ is the standard condition for the line to be a **tangent**.

Worked example. Find the condition for $y = mx + c$ to be tangent to $y = x^2$.

$$x^2 = mx + c \implies x^2 - mx - c = 0$$

Tangency requires $\Delta = 0$: $(-m)^2 - 4(1)(-c) = m^2 + 4c = 0$.

6. Simultaneous equations

Two linear equations: eliminate or substitute (a4). One linear and one quadratic: **always substitute the linear into the quadratic** (a5).

Worked example. Solve $y = x + 1$ and $x^2 + y^2 = 25$.

$$x^2 + (x + 1)^2 = 25 \implies 2x^2 + 2x + 1 = 25 \implies x^2 + x - 12 = 0$$

$$(x + 4)(x - 3) = 0 \implies x = -4 \text{ or } x = 3$$

Substituting back into $y = x + 1$: the solutions are $(-4, -3)$ and $(3, 4)$.

Substitute back into the **linear** equation, not the quadratic — it is quicker and cannot introduce spurious roots. And always pair the values: writing " $x = 3$ or $x = -4$, $y = 4$ or $y = -3$ " without pairing loses marks.

Geometrically the solutions are the **points of intersection** of the two graphs (a6). If the resulting quadratic has no real roots, the graphs do not meet.

7. Inequalities

Linear

Solve as an equation, but **reverse the inequality sign when multiplying or dividing by a negative**.

Quadratic

Factorise, find the critical values, then read the sign from a sketch (a8).

Worked example. Solve $x^2 - 4x - 5 > 0$.

$$(x - 5)(x + 1) > 0$$

Critical values $x = -1$ and $x = 5$. The parabola opens upwards, so it is *above* the axis outside the roots:

$$x < -1 \quad \text{or} \quad x > 5$$

A quick sketch settles the direction every time. For an upward parabola: < 0 between the roots, > 0 outside them. Guessing gets it backwards half the time.

Inequalities with an algebraic fraction

Do **not** multiply both sides by $x - 3$: its sign is unknown, so the inequality may or may not reverse. Instead collect on one side over a common denominator, or multiply by the *square* $(x - 3)^2$, which is always positive.

Set notation

Solutions may be expressed with "and"/"or" or in set notation (a9):

- $x \leq 1$ or $x \geq 4 \rightarrow \{x : x \leq 1\} \cup \{x : x \geq 4\}$
- $2 < x < 5 \rightarrow \{x : x > 2\} \cap \{x : x < 5\}$

Two separate intervals need **or** (union). Writing " $x < -1$ and $x > 5$ " describes numbers that are simultaneously less than -1 and greater than 5 — nothing at all.

8. Proportion

Statement	Equation	Graph
y directly proportional to x	$y = kx$	Straight line through the origin
y inversely proportional to x	$y = \frac{k}{x}$	Hyperbola
y proportional to x^n	$y = kx^n$	Depends on n
y inversely proportional to x^2	$y = \frac{k}{x^2}$	Inverse-square curve

Worked example. y is inversely proportional to x^2 , and $y = 4$ when $x = 3$. Find y when $x = 6$.

$$y = \frac{k}{x^2} \implies 4 = \frac{k}{9} \implies k = 36$$

$$x = 6 : \quad y = \frac{36}{36} = 1$$

9. Partial fractions

Partial fractions split a single algebraic fraction into a sum of simpler ones (spec ref a15). MEI restricts this to denominators with up to **three linear factors**, and includes **repeated (squared)**

linear factors.

Denominator	Form to assume
$(x + a)(x + b)$	$\frac{A}{x + a} + \frac{B}{x + b}$
$(x + a)(x + b)(x + c)$	$\frac{A}{x + a} + \frac{B}{x + b} + \frac{C}{x + c}$
$(x + a)^2(x + b)$	$\frac{A}{x + a} + \frac{B}{(x + a)^2} + \frac{C}{x + b}$

Worked example. Express $\frac{5x + 1}{(x + 1)(x - 2)}$ in partial fractions.

$$\frac{5x + 1}{(x + 1)(x - 2)} = \frac{A}{x + 1} + \frac{B}{x - 2}$$

Multiply through by $(x + 1)(x - 2)$:

$$5x + 1 = A(x - 2) + B(x + 1)$$

Substitute strategic values. Let $x = 2$: $11 = 3B$, so $B = \frac{11}{3}$. Let $x = -1$: $-4 = -3A$, so $A = \frac{4}{3}$.

$$\frac{5x + 1}{(x + 1)(x - 2)} = \frac{4}{3(x + 1)} + \frac{11}{3(x - 2)}$$

For a **repeated** factor $(x + a)^2$ you need *both* $\frac{A}{x + a}$ and $\frac{B}{(x + a)^2}$. Omitting the first term is the single most common error here — the system then has no solution and the algebra collapses.

Substituting the values that make each factor zero is fastest. If a term remains (as with a repeated factor), compare coefficients of the highest power, or substitute one more convenient value such as $x = 0$.

10. Common pitfalls

- Writing $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$.
- Forgetting the negative root: $x^2 = 16$ gives $x = \pm 4$, not just 4.
- Multiplying an inequality by a negative — or by an expression of unknown sign — without reversing or justifying.
- Joining two separate solution intervals with "and" instead of "or".
- Reading a quadratic inequality off the wrong side of the parabola.
- Omitting the $\frac{A}{x+a}$ term for a repeated linear factor in partial fractions.
- Confusing "two distinct real roots" ($\Delta > 0$) with "real roots" ($\Delta \geq 0$).
- Losing the pairing of x and y values in simultaneous equations.